Importance-Weight Guidance in Masked Block Diffusion — Formulation & Kernel Analysis

Adapting the discriminator-guided reverse process of *Controllable Graph Diffusion Path Generation* (Eqs. 15–36 of the draft) from the uniform-state CTMC kernel to the absorbing (masked) kernel. Goal: train on the *normal* graph and steer generation toward an unseen *exceptional* distribution q using a discriminator D_ϕ, by importance-reweighting the model's $\hat{x}_0$ proposals.

1. Setup and notation

A path is $x \in \{0,1\}^{L\times K}$; row x^l is the one-hot state at position l over K states. q = target (exceptional) data distribution; p_θ = model trained on normal data; $D_\phi(x_0) \approx \Pr[\text{exceptional} \mid x_0]$. We denote by $\hat{x}_0^{\,l} = [\,p_\theta(x_0^l = k \mid x_t)\,]_k$ the per-position x_0-prediction. Two within-block kernels:

- **Uniform (CTMC)**: $Q_t = \exp((A-D)t)$, cumulative $\bar{Q}_t = \prod_{i\le t} Q_i$; posterior $q(x_{t-1}^l \mid x_t^l, x_0^l) = \mathrm{Cat}\big(p \propto x_t^l Q_t^\top \odot x_0^l \bar{Q}_{t-1}\big)$.
- **Absorbing (mask)**: extra state $m = \langle \mathrm{MASK} \rangle$; survival prob α_t ($\alpha_0 = 1,\ \alpha_T \to 0$), $q(x_t^l = x_0^l) = \alpha_t,\ q(x_t^l = m) = 1 - \alpha_t$ (defined in §4).

2. Target reverse posterior via importance weighting (recap)

The exceptional-target reverse posterior is rewritten as an expectation under the model's x_0-prediction and importance-corrected (draft Eqs. 16–20):

$$q(x_{t-1} \mid x_t) = \mathbb{E}_{x_0 \sim p_\theta(x_0|x_t)}\Big[\tfrac{q(x_0|x_t)}{p_\theta(x_0|x_t)}\, q(x_{t-1} \mid x_t, x_0)\Big] \;\approx\; \tfrac{1}{Z}\sum_{n=1}^{N} \tfrac{q(x_0^{(n)})}{p_\theta(x_0^{(n)})}\, q(x_{t-1} \mid x_t, x_0^{(n)}) \tag{1}$$

Using the **same forward corruption** for q and p_θ, the factor $q(x_t \mid x_0)/p_\theta(x_t \mid x_0) = 1$, and under $p_\theta = q_{\text{normal}}$ the weight collapses to a discriminator density ratio (draft Eqs. 27–29):

$$\frac{q(x_0 \mid x_t)}{p_\theta(x_0 \mid x_t)} \;\propto\; \frac{q(x_0)}{p_\theta(x_0)} \;=\; \frac{D_\phi(x_0)}{1 - D_\phi(x_0)} \tag{2}$$

Because both the true and predicted posteriors factorize over positions, the guided reverse step is a product of per-position categoricals with the reweighted target $\bar{x}_0$ in place of $\hat{x}_0$ (draft Eq. 30):

$$q(x_{t-1} \mid x_t) \approx \bigotimes_{l=1}^{L} \mathrm{Cat}\big(x_{t-1}^l;\, p \propto \underbrace{q(x_{t-1}^l \mid x_t^l, x_0^l = \bar{x}_0^{\,l})}_{\text{kernel-specific}}\big), \qquad \bar{x}_0^{\,l} = \Big[\textstyle\sum_{n=1}^{N} \mathbf{1}\{x_0^{(n),l} = k\}\, \frac{D_\phi(x_0^{(n)})}{1 - D_\phi(x_0^{(n)})}\Big]_k \tag{3}$$

3. Kernel-agnostic vs. kernel-specific

Only **one** ingredient of (1)–(3) is kernel-specific.

ingredient	holds for mask?
position factorization $q(x_{t-1} \mid x_t, x_0) = \prod_l q(x_{t-1}^l \mid x_t^l, x_0^l)$	Yes — absorbing forward is position-independent
cancellation $q(x_t \mid x_0)/p_\theta(x_t \mid x_0) = 1$	Yes — same absorbing forward for q, p_θ
ratio $q(x_0)/p_\theta(x_0) = D_\phi/(1 - D_\phi)$	Yes — defined on clean paths, kernel-free
product-form $\hat{x}_0 = \prod_l p_\theta(x_0^l \mid x_t)$	Yes — SUBS also per-position
closed form of $q(x_{t-1}^l \mid x_t^l, x_0^l)$	**No — this is the only piece to replace**

So the guidance transfers by swapping the CTMC posterior for the absorbing posterior below; the reweighting machinery is unchanged.

4. Absorbing-kernel instantiation

Forward (per position): $q(x_t^l = x_0^l \mid x_0^l) = \alpha_t, \quad q(x_t^l = m \mid x_0^l) = 1 - \alpha_t$. The true posterior is (standard absorbing form):

$$q(x_{t-1}^l \mid x_t^l, x_0^l) = \begin{cases} \delta(x_{t-1}^l = x_t^l) & x_t^l \neq m \ \ (\text{revealed} \Rightarrow \text{frozen}) \\ \dfrac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t}\, \delta(x_{t-1}^l = x_0^l) + \dfrac{1 - \alpha_{t-1}}{1 - \alpha_t}\, \delta(x_{t-1}^l = m) & x_t^l = m \end{cases} \tag{4}$$

Guided step: substitute the reweighted $\bar{x}_0^{\,l}$ of (3) as the reveal-target at masked positions. Concretely, a masked token is revealed to state k with probability $\dfrac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t}\, \bar{x}_0^{\,l}[k]$, else stays m:

$$q(x_{t-1}^l \mid x_t^l = m) \;\approx\; \frac{\alpha_{t-1} - \alpha_t}{1 - \alpha_t}\, \bar{x}_0^{\,l} \;+\; \frac{1 - \alpha_{t-1}}{1 - \alpha_t}\, \mathbf{e}_m \tag{5}$$

Only currently-masked positions are steerable; revealed tokens are frozen (copy-through). Under first-hitting sampling this means the reweighting acts on the single token being revealed per step, using D_ϕ of $[\text{committed prefix} \,\|\, \text{candidate completion}]$.

---

5. The marginal-target correction (draft §2.1) degenerates under masking

Draft §2.1 (LiDAR) draws candidates from the iterative marginal $x_0^{(n)} \sim p_\theta(x_0)$ and corrects with a forward-kernel weight, giving $\bar{x}_0^{\,l} = \big[\sum_n \mathbf{1}\{x_0^{(n),l} = k\}\, w_n \,\frac{D_\phi}{1-D_\phi}\big]_k$ with $w_n = \dfrac{q(x_t \mid x_0^{(n)})}{\frac{1}{N}\sum_i q(x_t \mid x_0^{(i)})}$. For the absorbing kernel, $q(x_t \mid x_0)$ depends only on the **mask pattern**, not on masked-position values:

$$q(x_t \mid x_0^{(n)}) = \prod_{l:\, x_t^l \neq m} \alpha_t \,\mathbf{1}\{x_0^{(n),l} = x_t^l\} \prod_{l:\, x_t^l = m} (1-\alpha_t) = \alpha_t^{|\mathrm{rev}|}(1-\alpha_t)^{\,|\mathrm{mask}|} \cdot \mathbf{1}\{x_0^{(n)} \text{ consistent on revealed } l\} \tag{6}$$

> **Consequence.** Among candidates consistent with the revealed positions, $q(x_t \mid x_0^{(n)})$ is **identical**; for inconsistent candidates it is 0. Hence w_n collapses to a **hard consistency indicator** (1 for consistent, 0 otherwise) and carries **no smooth information**. The §2.1 estimator therefore reduces to Eq. 30 (plain $D/(1-D)$ weighting) in the mask kernel:
>
> $$w_n \xrightarrow{\text{absorbing}} \mathbf{1}\{x_0^{(n)} \text{ consistent}\}, \qquad \bar{x}_0^{\,l} \;\to\; \Big[\textstyle\sum_{n:\,\text{consistent}} \mathbf{1}\{x_0^{(n),l} = k\}\,\frac{D_\phi}{1-D_\phi}\Big]_k \tag{7}$$

In the uniform kernel the CTMC forward assigns genuinely different $q(x_t \mid x_0^{(n)})$ to different x_0 (non-absorbing), so w_n is informative — the §2.1 refinement's advantage over the one-step baseline is thus **largely uniform-kernel-specific**.

---

6. Practical bottleneck: proposal entropy (not the algebra)

The estimator $\bar{x}_0$ differs from $\hat{x}_0$ only if the N candidates $x_0^{(n)} \sim p_\theta(x_0 \mid x_t)$ are **diverse** (so distinct $D/(1-D)$ weights reshape the average). The mask kernel's infilling is typically low-entropy (confident given context) $\Rightarrow$ near-clone candidates $\Rightarrow \bar{x}_0 \approx \hat{x}_0 \Rightarrow$ guidance is inert. This matches the empirical finding that reweighting is flat on the mask kernel at every λ/block size/discriminator (incl. oracle), while it doubles validity on the higher-entropy uniform kernel.

Remedies are proposal-side, not derivational: (i) raise candidate sampling temperature; (ii) draw candidates under the exceptional scenario's adjacency (adjacency-masked proposals); (iii) brief proposal fine-tuning on exceptional data. Alternatively, keep the uniform kernel — whose soft proposals make guidance effective and whose adjacency-constrained decode gives validity on unseen graphs for free — and address its normal-case quality via respaced sampling.

---

7. Summary

- The importance-weighted guidance (Eqs. 1–3) is **kernel-agnostic except for** $q(x_{t-1}^l \mid x_t^l, x_0^l)$; substituting the absorbing posterior (4)–(5) yields a valid masked-diffusion guidance scheme.
- The §2.1 forward-kernel correction **degenerates to a consistency indicator** under masking (Eqs. 6–7), reducing to the plain $D/(1-D)$ estimator; its refinement is meaningful mainly for the uniform kernel.
- The binding constraint is **proposal entropy**: masked infilling is too confident for reweighting to act, so the fix is proposal-side (temperature / adjacency-masked proposals / fine-tune), or use the uniform kernel and fix its quality via respaced sampling.